

CNN Models for Computer Vision - Transfer Learning and Fine Tuning

Advanced Computer Vision Techniques

PhD Program in Artificial Intelligence
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1. Test the “Recent_Models.ipynb” for each the general models under consideration (VGG, ResNet, Inception)
2. Create a full Pipeline for selecting the best configuration for each model. Use a Random Search hyperparameter tuning for learning rate, batch size, optimizer, and regularization approach
3. Report each *best model* performance using the proper metrics for Testing
4. Estimate and report and analyze the performance of each method for:
 - a) Batch (step) and Epoch
 - b) Training loss convergence
 - c) Classification performance on CIFAR 10
5. Adapt the resulting transfer learning and fine tuning framework to perform the same experimentation on another multiclass, multichannel, “small” image dataset
6. Qualitatively analyze some representative examples on the posthoc interpretability provided by GradCAM